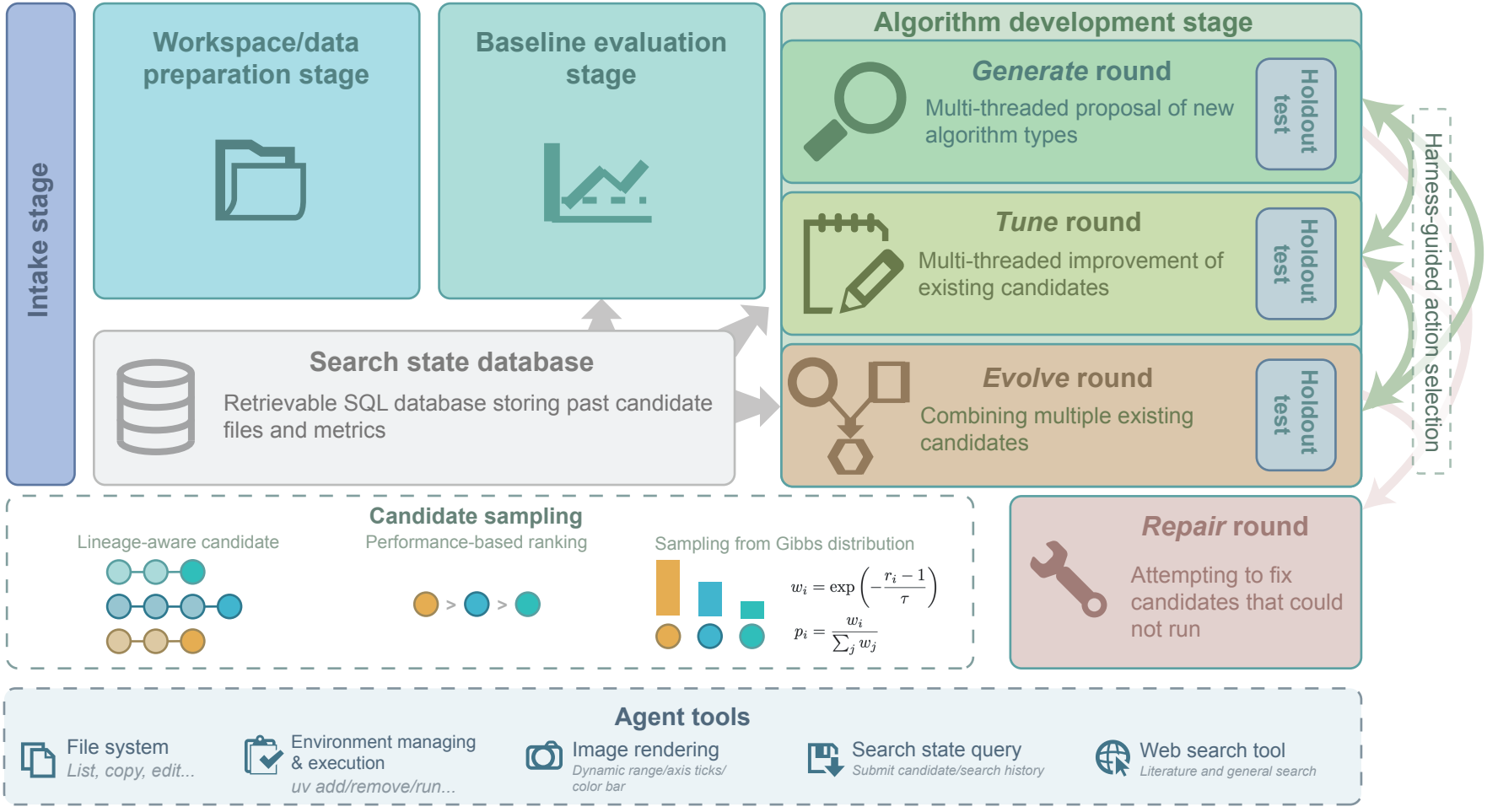

Intake stage
Workspace/data preparation stage
Baseline evaluation stage
Algorithm development stage
Generate round
Multi-threaded proposal of new algorithm types
Holdout test
Tune round
Multi-threaded improvement of existing candidates
Holdout test
Evolve round
Combining multiple existing candidates
Holdout test
Harness-guided action selection
Search state database
Retrievable SQL database storing past candidate files and metrics
Candidate sampling
Lineage-aware candidate
Performance-based ranking
Sampling from Gibbs distribution
$w_i = \exp\left(-\frac{r_i - 1}{\tau}\right)$
$p_i = \frac{w_i}{\sum_j w_j}$
Repair round
Attempting to fix candidates that could not run
Agent tools
File system
List, copy, edit...
Environment managing & execution
uv add/remove/run...
Image rendering
Dynamic range/axis ticks/ color bar
Search state query
Submit candidate/search history
Web search tool
Literature and general search